

Generalized measurement system

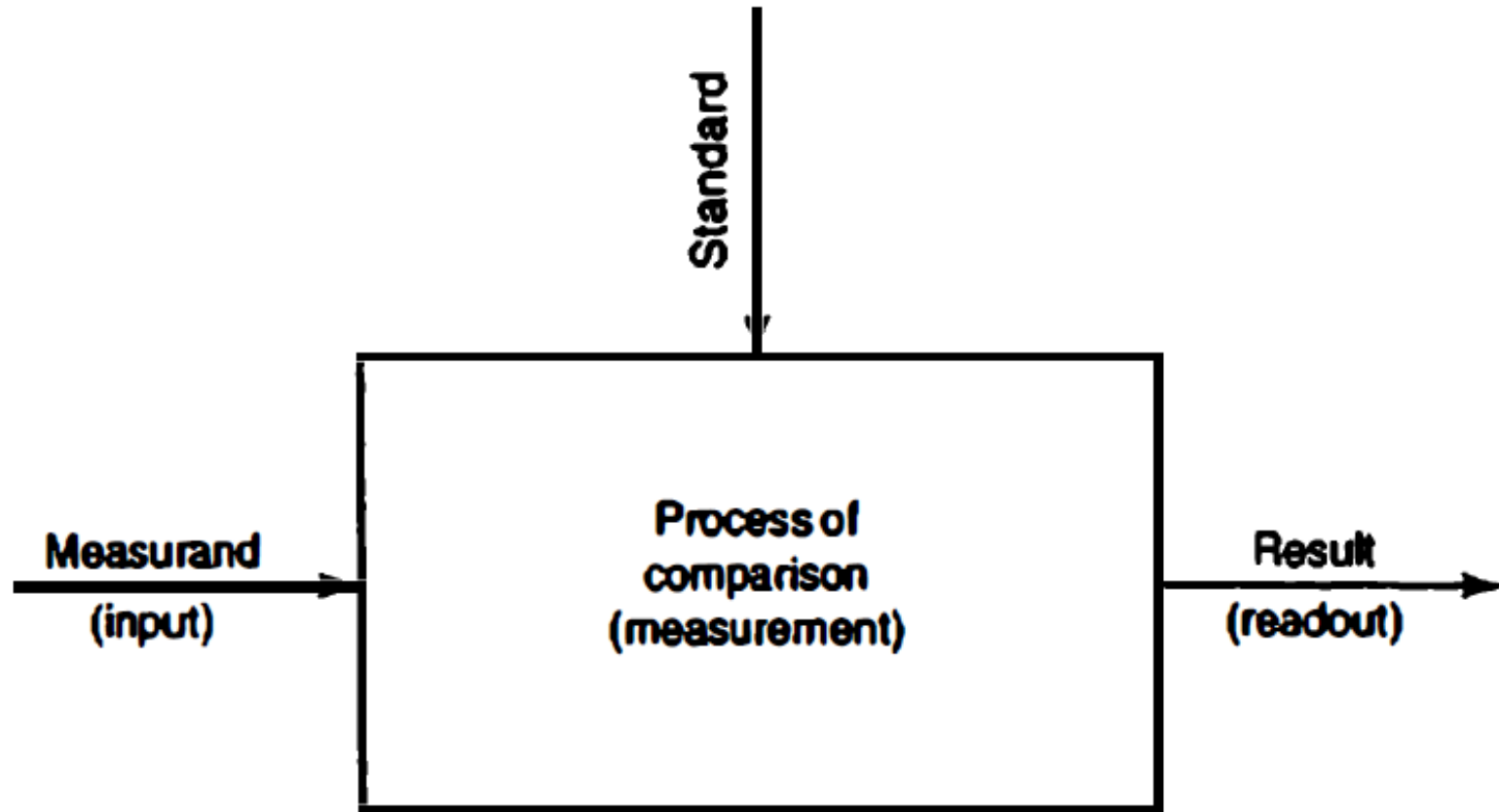
Measurand

- What you want to measure :
 Measurands (e.g. temperature;
 pressure; speed, ...)

Measurand

- The word measurand is used to designate the particular physical parameter being observed , i.e the unknown quantity which is to be measured.
- It is the input quantity to the measuring process
- This unknown quantity is compared with the available standard quantities.

Fundamental measuring process



- 1- Sensing element:** it has a physical characteristic that responds to the variation of the measurand.
- 2- Signal modification subsystem:** it will modify (without altering) the output (signal) of the sensing element to make it more suitable for reading or recording (e.g. amplification, filtering, averaging, conversion to physical units, ...).
- 3- Indicator or recorder:** it will display the output (usually in physical units of measurand) and/or record it.

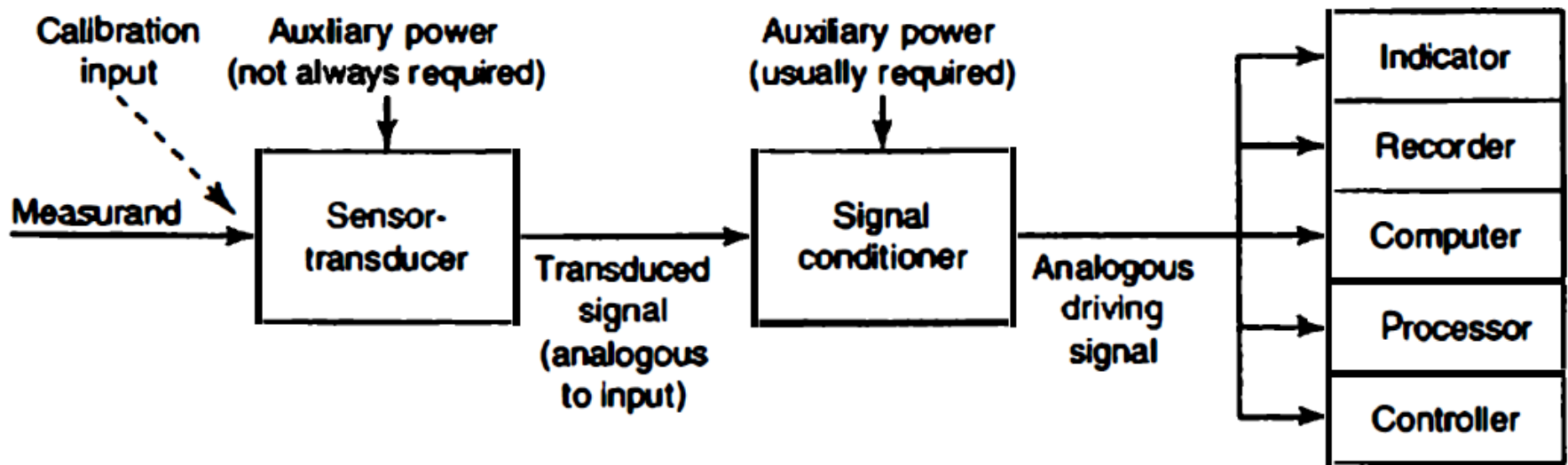
THE GENERALIZED MEASURING SYSTEM

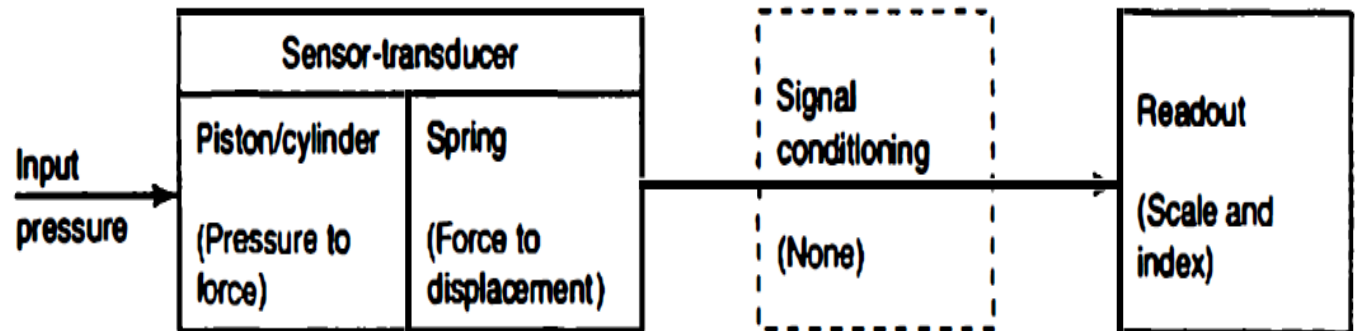
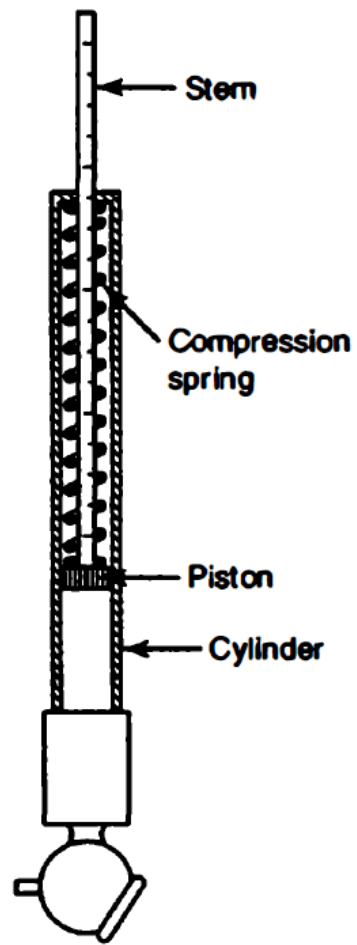
Stage 1. A detection-transduction, or sensor-transducer, stage

Stage 2. An intermediate stage, which we shall call the signal-conditioning stage

Stage 3 • . A terminating, or readout-recording stage

Block diagram of the generalized measuring system





Static Characteristics of Instruments

If we have a thermometer in a room and its reading shows a temperature of 20°C , then it does not really matter whether the true temperature of the room is 19.5°C or 20.5°C . Such small variations around 20°C are too small to affect whether we feel warm enough or not. Our bodies cannot discriminate between such close levels of temperature and therefore a thermometer with an inaccuracy of 0.5°C is perfectly adequate.

If we had to measure the temperature of certain chemical processes, however, a variation of $0.5\text{ }^{\circ}\text{C}$ might have a significant effect on the rate of reaction or even the products of a process. A measurement inaccuracy much less than $0.5\text{ }^{\circ}\text{C}$ is therefore clearly required.

Accuracy of measurement is thus one consideration in the choice of instrument for a particular application. Other parameters, such as sensitivity, linearity, and the reaction to ambient temperature changes, are further considerations. These attributes are collectively known as the static characteristics of instruments and are given in the data sheet for a particular instrument.

Accuracy and Inaccuracy (Measurement Uncertainty)

The accuracy of an instrument is a measure of how close the output reading of the instrument is to the correct value. In practice, it is more usual to quote the inaccuracy or measurement uncertainty value rather than the accuracy value for an instrument.

Inaccuracy or measurement uncertainty is the extent to which a reading might be wrong and is often quoted as a percentage of the full-scale (f.s.) reading of an instrument.

Precision

Precision is a term that describes an instrument's degree of freedom from random errors.

If a large number of readings are taken of the same quantity by a high-precision instrument, then the spread of readings will be very small.

Precision is often, although incorrectly, confused with accuracy. High precision does not imply anything about measurement accuracy. A high-precision instrument may have a low accuracy. Low accuracy measurements from a high-precision instrument are normally caused by a bias in the measurements, which is removable by recalibration.

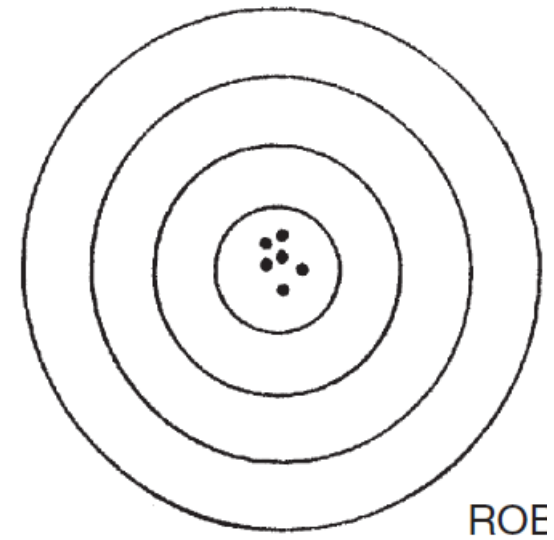
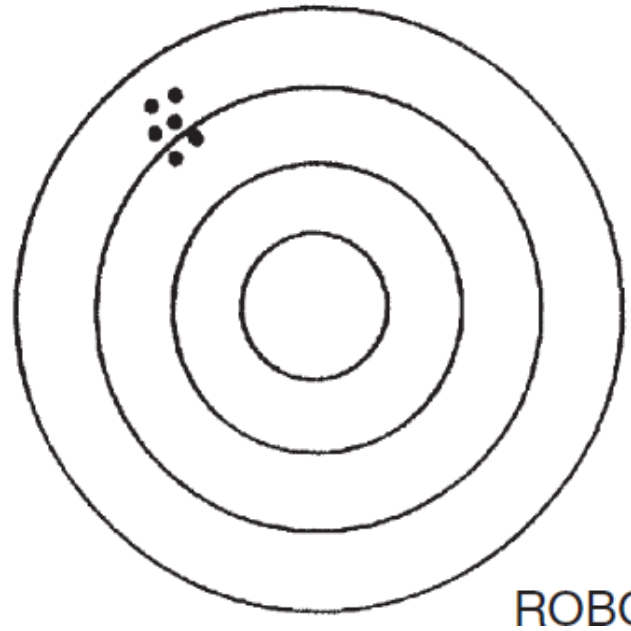
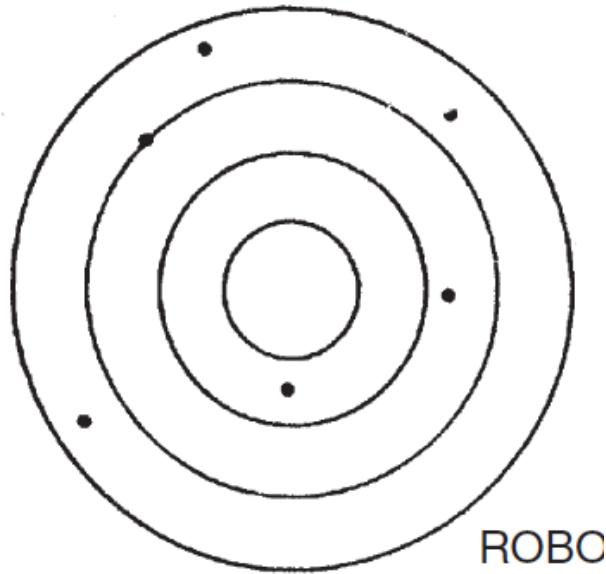
Repeatability / Reproducibility

The terms repeatability and reproducibility mean approximately the same but are applied in different contexts, as given later.

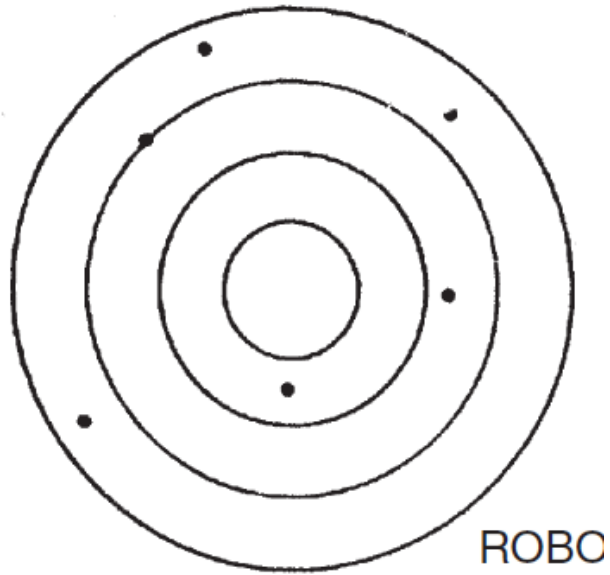
Repeatability describes the closeness of output readings when the same input is applied repetitively over a short period of time, with the same measurement conditions, same instrument and observer, same location, and same conditions of use maintained throughout.

Reproducibility describes the closeness of output readings for the same input when there are changes in the method of measurement, observer, measuring instrument, location, conditions of use, and time of measurement. Both terms thus describe the spread of output readings for the same input. This spread is referred to as repeatability if the measurement conditions are constant and as reproducibility if the measurement conditions vary.

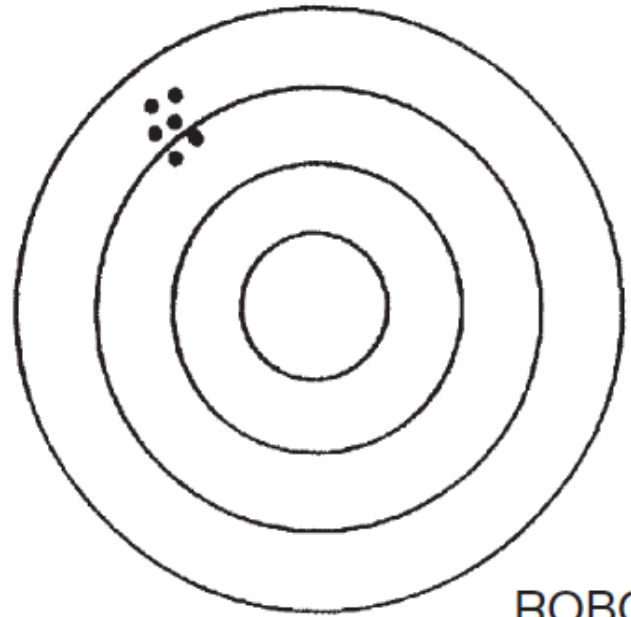
Comparison of accuracy and precision



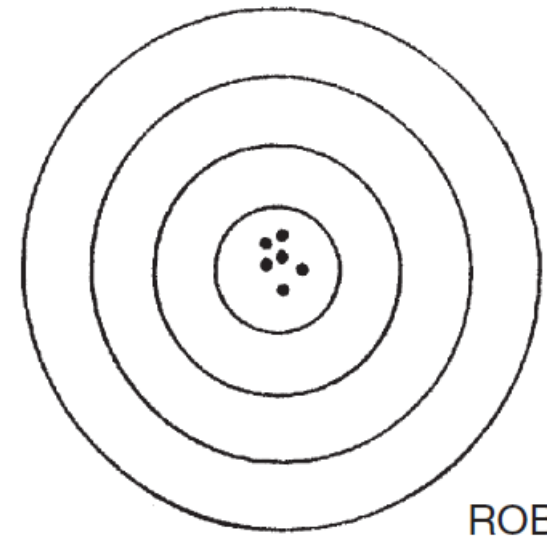
Comparison of accuracy and precision



(a) Low precision,
low accuracy



(b) High precision,
low accuracy



(c) High precision,
high accuracy

Tolerance

Tolerance is a term that is closely related to accuracy and defines the maximum error that is to be expected in some value.

While it is not, strictly speaking, a static characteristic of measuring instruments, it is mentioned here because the accuracy of some instruments is sometimes quoted as a tolerance value.

When used correctly, tolerance describes the maximum deviation of a manufactured component from some specified value.

For instance, crankshafts are machined with a diameter tolerance quoted as so many micrometers (10^{-6} m), and electric circuit components such as resistors have tolerances of perhaps 5%.

Range or Span

The range or span of an instrument defines the minimum and maximum values of a quantity that the instrument is designed to measure.

Linearity

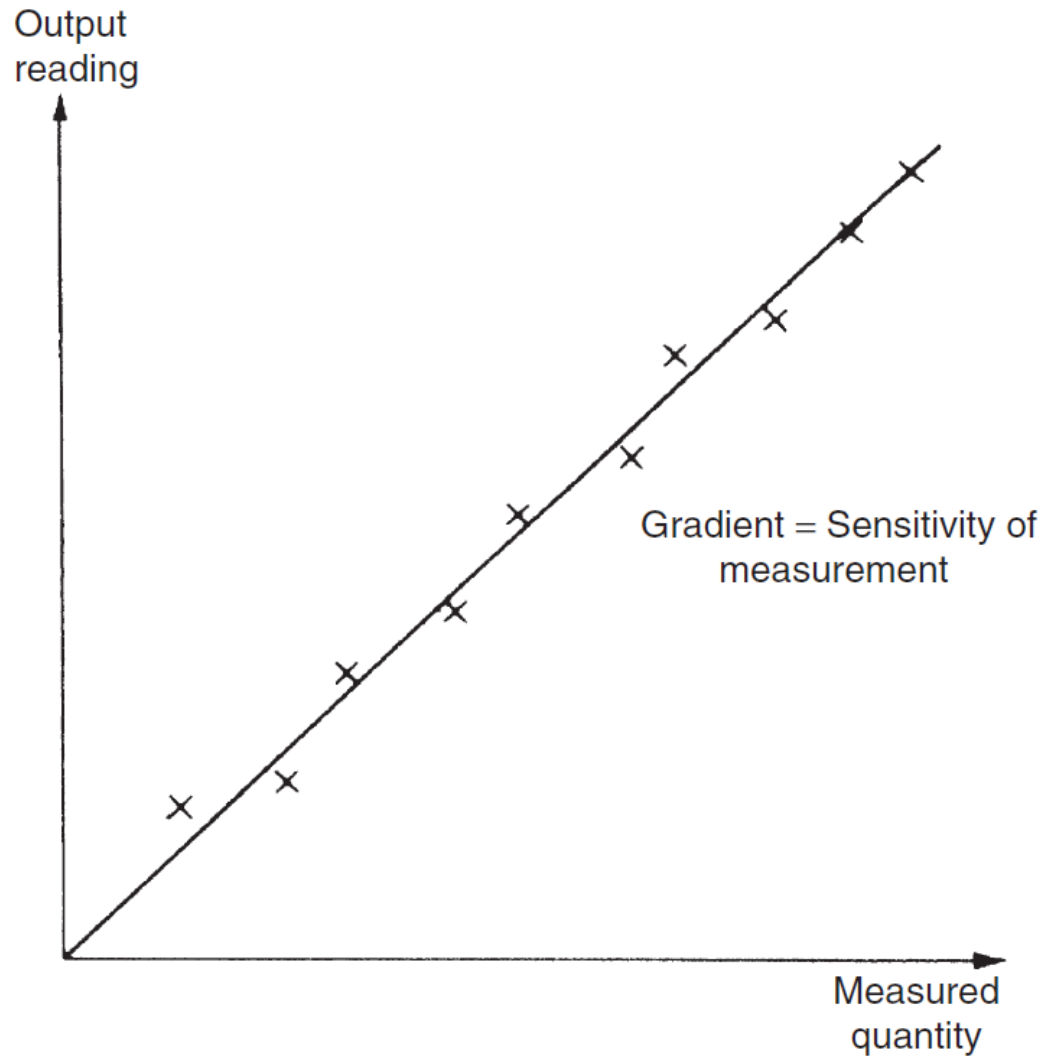
It is normally desirable that the output reading of an instrument is linearly proportional to the quantity being measured. The Xs marked on Figure show a plot of typical output readings of an instrument when a sequence of input quantities are applied to it. Normal procedure is to draw a good fit straight line through the Xs, as shown in Figure.

(While this can often be done with reasonable accuracy by eye, it is always preferable to apply a mathematical least-squares line-fitting technique.)

Nonlinearity is then defined as the maximum deviation of any of the output readings marked X from this straight line.

Nonlinearity is usually expressed as a percentage of full-scale reading.

Instrument output characteristic



Sensitivity of Measurement

The sensitivity of measurement is a measure of the change in instrument output that occurs when the quantity being measured changes by a given amount.

Thus, sensitivity is the ratio:

$$\frac{\text{scale deflection}}{\text{value of measurand producing deflection}}$$

The sensitivity of measurement is therefore the slope of the straight line drawn on Figure.

If, for example, a pressure of 2 bar produces a deflection of 10 degrees in a pressure transducer, the sensitivity of the instrument is 5 degrees/bar (assuming that the deflection is zero with zero pressure applied).

Threshold

If the input to an instrument is increased gradually from zero, the input will have to reach a certain minimum level before the change in the instrument output reading is of a large enough magnitude to be detectable. This minimum level of input is known as the threshold of the instrument.

As an illustration, a car speedometer typically has a threshold of about 15 km/h. This means that, if the vehicle starts from rest and accelerates, no output reading is observed on the speedometer until the speed reaches 15 km/h.

Applications of Instrument Systems

- Measurement of system parameters informations

For example, a pressure gauge fitted on a steam boiler drum gives the visual indication of pressure of steam. Further, the deflection of pointer of speedometer monitors the speed of automobile at that moment.

- **Control of a certain process or operation**

For example, a thermostat fitted in a domestic refrigerator is a control device for maintaining the temperature in a specified range.

Currently, automatic control systems are widely used in process industries like oil refineries, chemical plants, textile mills, etc. for controlling variables like temperature, pressure, humidity, viscosity, flow rate and other relevant parameters. Furthermore, they are also used in modern sophisticated systems like autopilots, automatic landing of aircraft, missile guidance, radar tracking systems, etc.

- **Simulation of system conditions**

Sometimes, it may be necessary to simulate experimentally the actual conditions of complex situations for revealing the true behaviour of the system under different governing conditions. The lift, drag and other relevant parameters of aerodynamic bodies are usually obtained by testing the models in controlled air streams generated in wind tunnels that simulate the flow conditions experienced by aerodynamic bodies. The information thus obtained is used in the design and development of the prototype.

- **Experimental design studies**

For example, a design team of experienced aircraft designers put in a number of years of effort to produce a prototype aircraft. The prototype is flown by a test pilot to determine the various performance/operating parameters. The prototype test data is then used to improve further the design calculations and a modified prototype is produced. This is carried on till the desired design performance is achieved.

- To perform various manipulations

In a number of cases, the instruments are employed to perform operations like signal addition, subtraction, multiplication, division, differentiation, integration, signal linearisation, signal sampling, signal averaging, multi-point correlations, ratio controls, etc.

In certain cases, instruments are also used to determine the solution of complex differential equations or other mathematical manipulations. A simple pocket calculator is an example of a mathematical processing instrument, to some extent.

- **Testing of materials, maintenance of standards and specifications of products**

Most countries have standards organisations that specify material standards and product specifications based on extensive tests and measurements. These organisations are meant to protect the interests of consumers. They ensure that the material/products meet the specified requirements so that they function properly and enhance the reliability of the system. For example, an aircraft engine is subjected to extensive endurance tests by the civil aviation authorities as per their specifications, before it is certified to be airworthy.

- Verification of physical phenomena/scientific theories

Coulomb postulated that the friction between two dry surfaces is proportional to the normal reaction and is independent of the area of contact. His hypothesis has since been verified experimentally and is now known as Coulomb's law of dry friction. In fact, such examples are numerous.

Whenever a scientist or an engineer proposes any hypothesis predicting the system's behaviour, it needs to be checked experimentally to put the same on a sound footing.

In addition, experimental studies play an important role in formulating certain empirical relations where adequate theory does not exist. For example, a number of empirical relations for the friction factor of turbulent flow in pipes (where theoretical basis is inadequate) have been formulated till date by various investigators based on their hypotheses in which numerical constants have been evaluated from experimental data.

- Quality control in industry

For example, a boiler plate has to undergo a number of quality control tests before it is put in actual operation.

Accuracy

- Accuracy is defined as the closeness of the measured value with true value.

Or

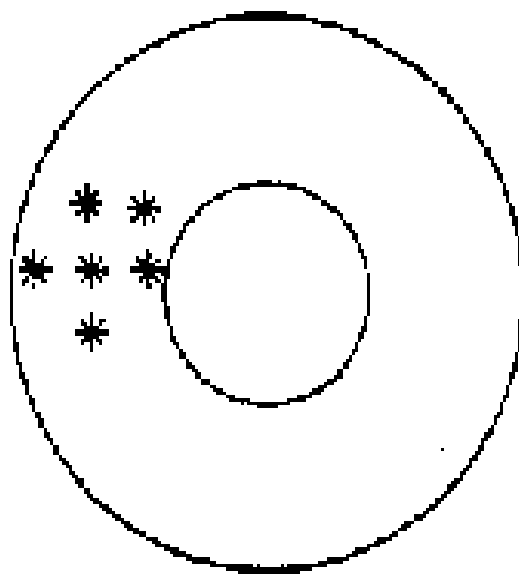
Accuracy is defined as the degree to which the measured value agrees with the true value

Precision

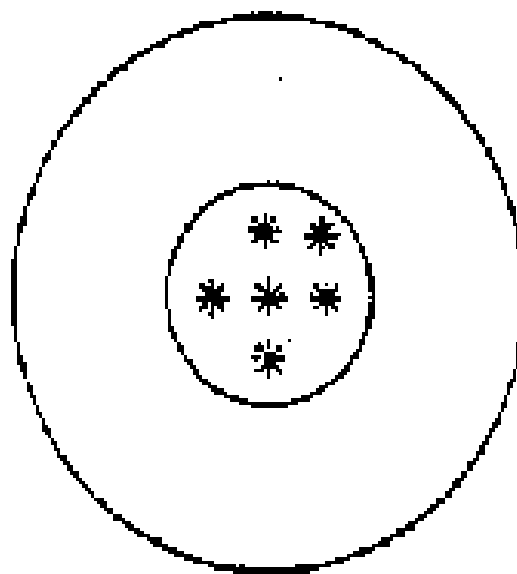
- A measure of how close repeated trials are to each other

Or

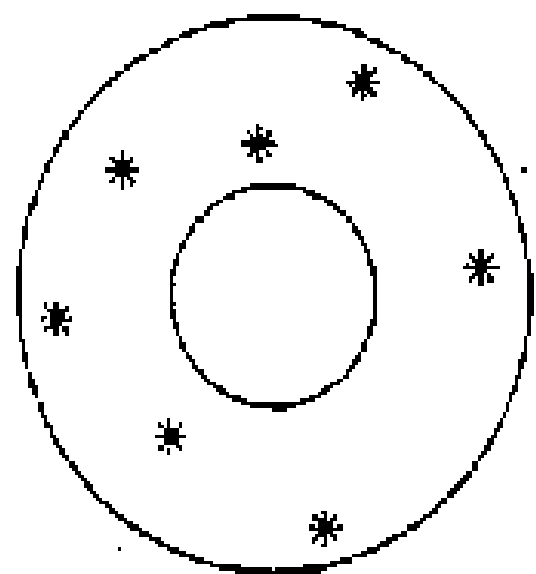
The closeness of repeated measurements



High precision
poor accuracy

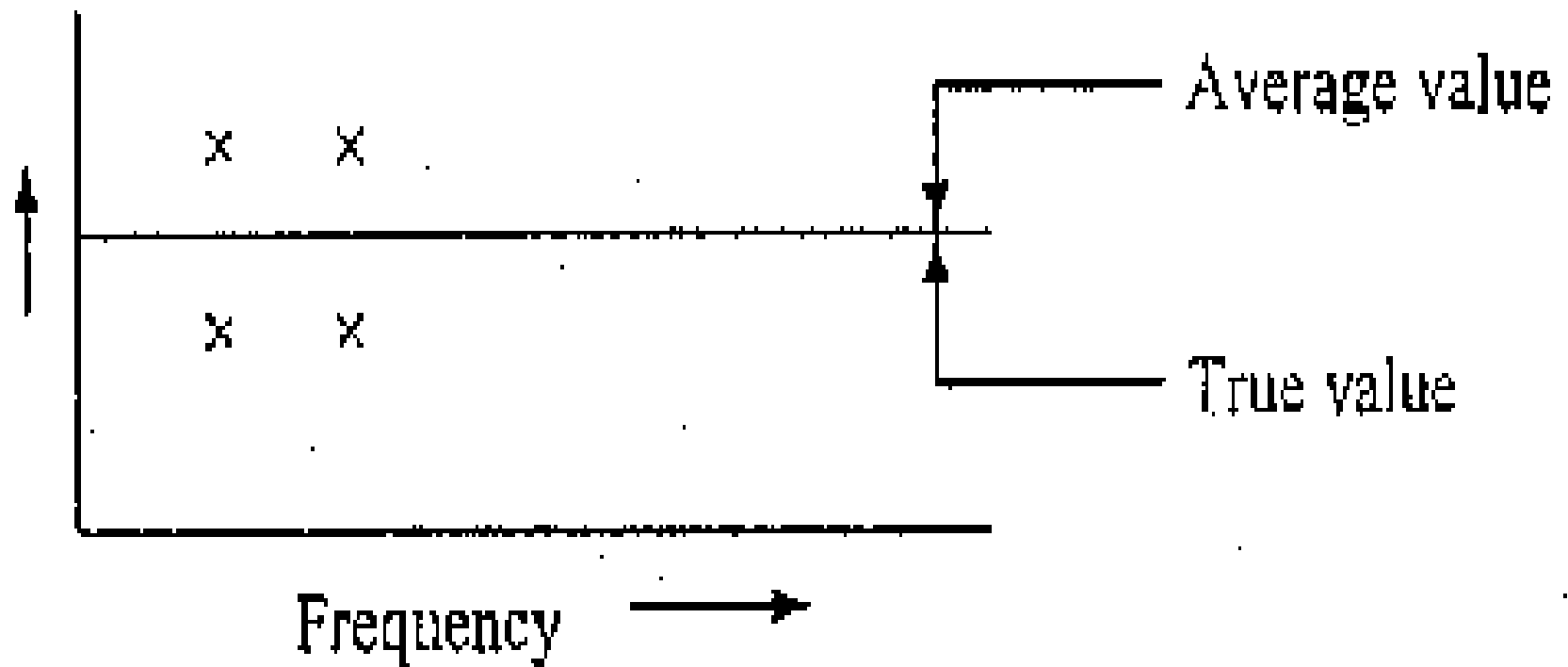


High precision
High accuracy

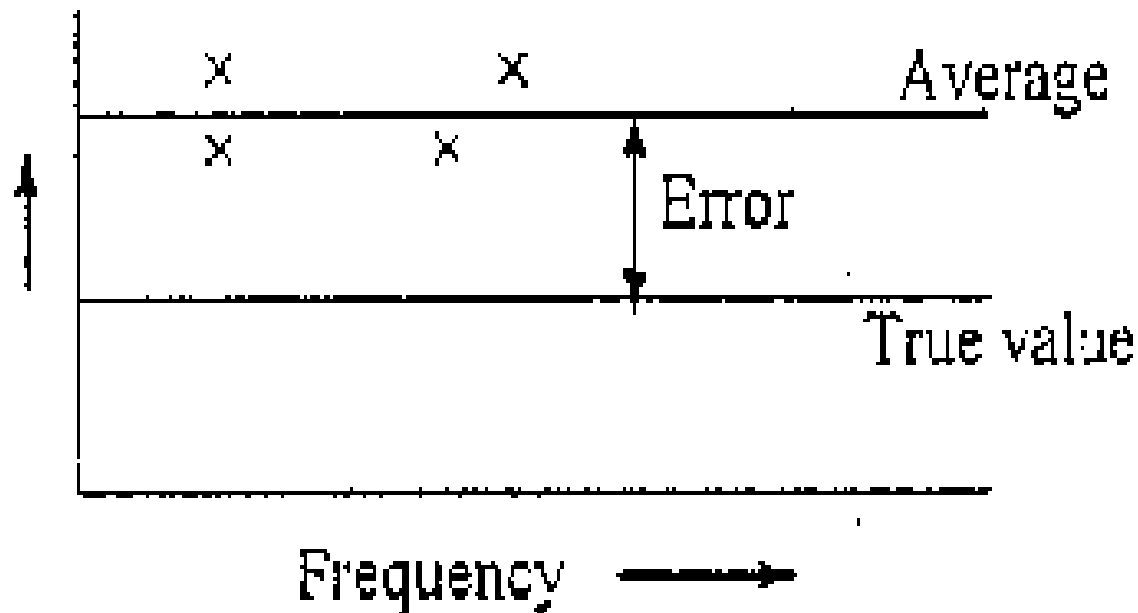


Poor precision
Good average accuracy

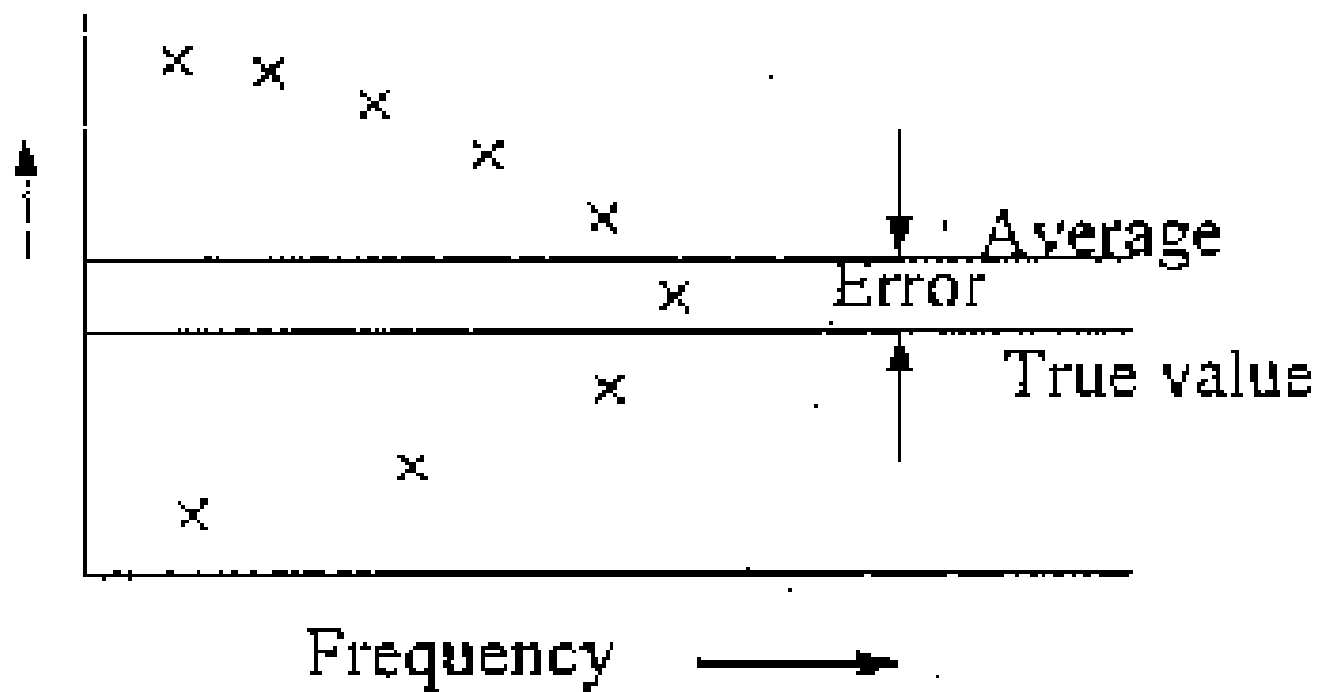
Fig. 1.11. Accuracy vs Precision



a) Accurate and precise



b) Precise but not Accurate (Results of measurement close to Average value)



c) Accurate but not precise

Error

- The error in the measurement is the difference between the measured value and the true value of measured dimensions

Error

$$\text{Error} = \text{MEASURED VALUE} - \text{TRUE VALUE}$$

- Absolute error
 - True absolute error
 - Apparent absolute error
- Relative error

True absolute error

- It is the algebraic difference between the result of measurement and the conventional true value of the quantity.

Apparent absolute error

- If the series of measurement are made then the algebraic difference between one of the results of measurement and the arithmetical mean is known as apparent absolute error

Relative error

- It is the quotient of absolute error and the true value or the arithmetical mean for series of measurement

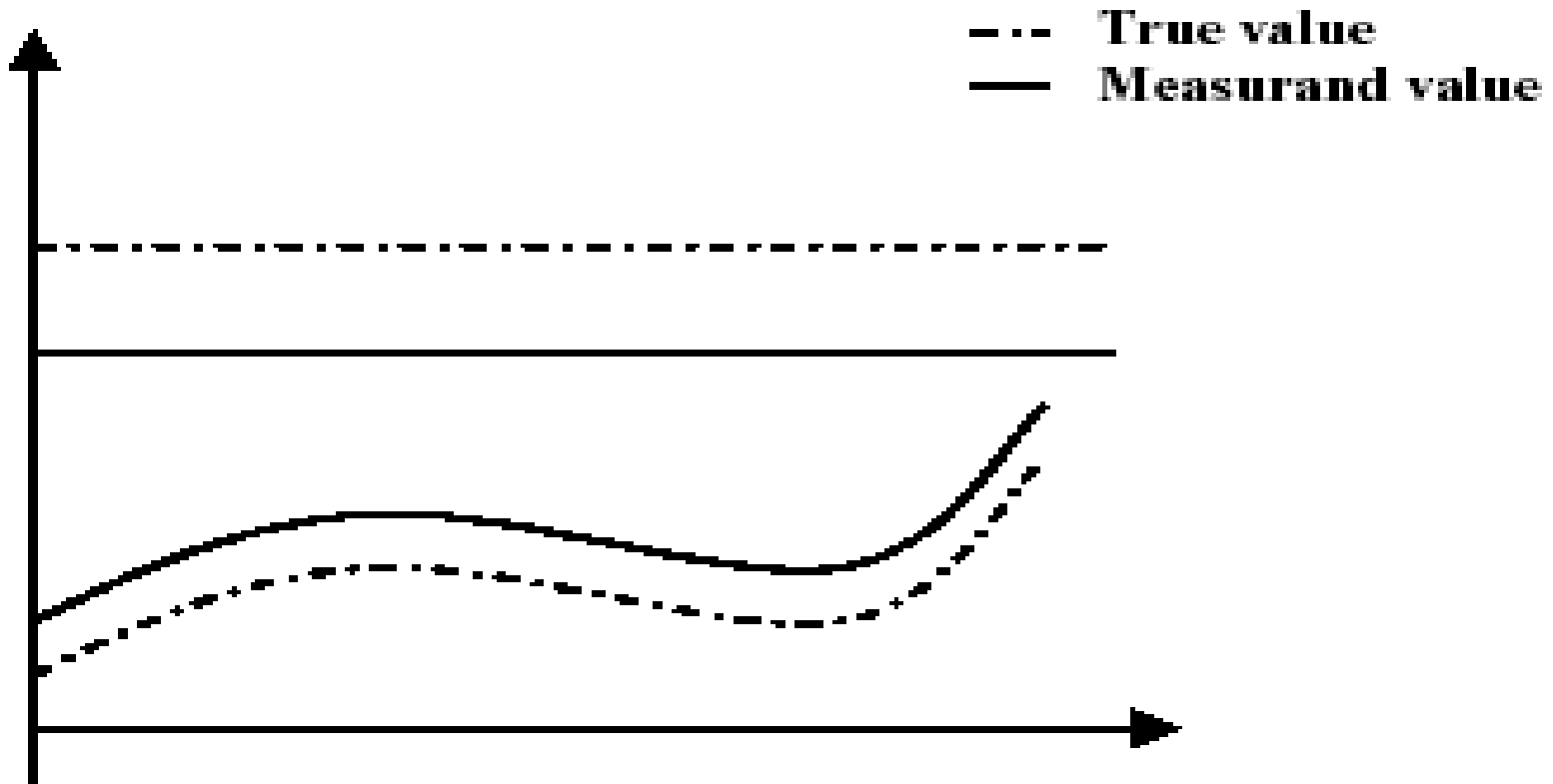
Types of Error

- Systematic error(Fixed or bias errors)
- Random error (Precision errors)

Systematic error(Fixed or bias errors)

- Systematic errors are consistent, keeping always the same value.

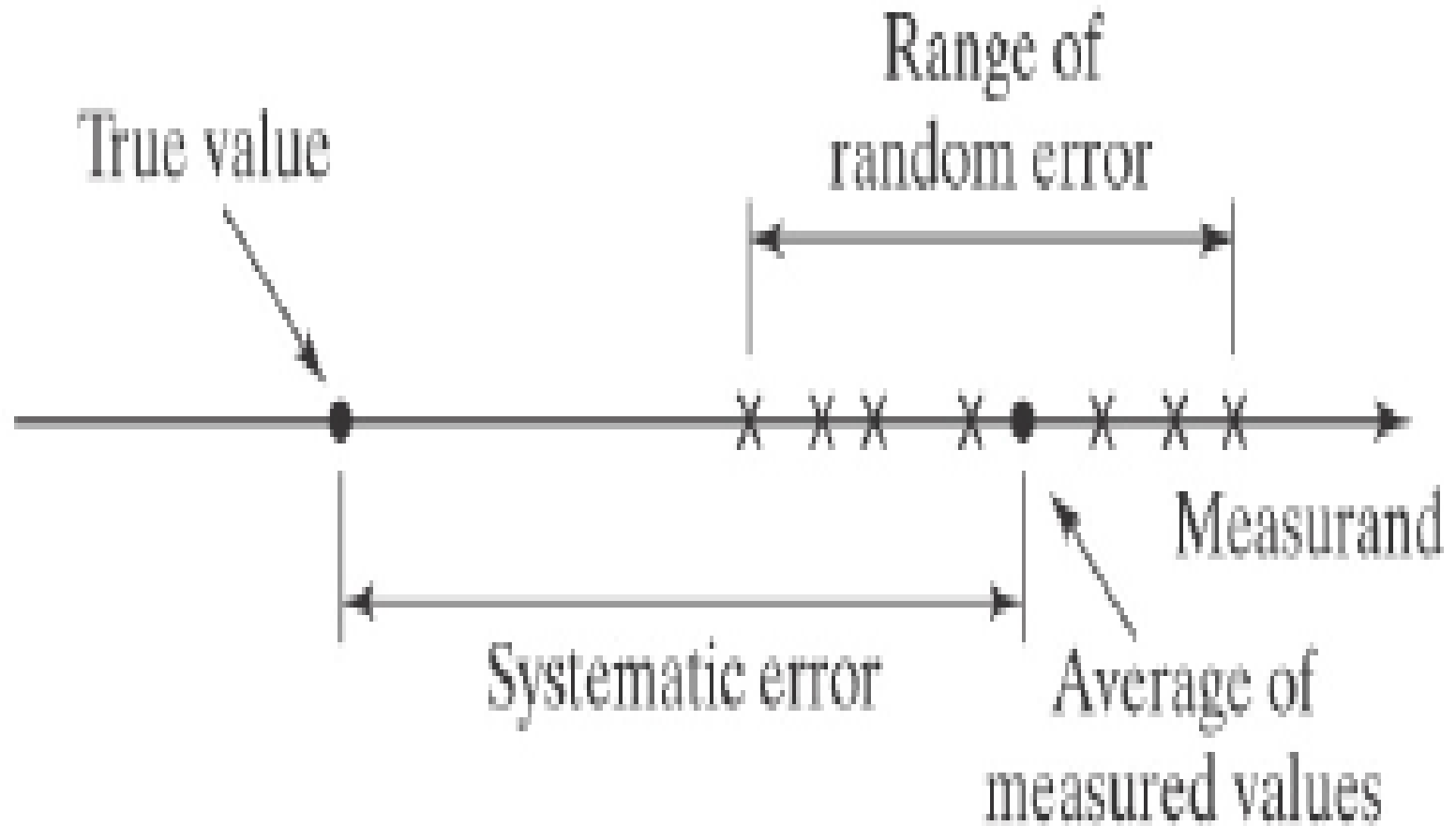
Systematic errors



Random error

- Random errors by definition are unpredictable and varying from one measurement to the other.
- This is why in order to determine an estimate of random errors, we need to record the measurand several times.

Random error



Systematic errors

- Instrument Errors
- the use of uncalibrated instruments
- Environment Errors

Random error

- Unexplained & unpredictable errors
- Measurements taken by human observation of analog meters
- Electrical noise

<i>Sl. No.</i>	<i>Systematic errors</i>	<i>Random errors</i>
1	It can be controlled by magnitude and sense.	It cannot be determined from the knowledge of measuring system.
2.	It is repetitive in nature.	It is non consistent
3.	Property analyzed can be determined and reduced.	Cannot be eliminated.
4.	These types of errors are due to improper conditions or procedures.	Random errors are inherent in the measuring system.
5.	These include the variation in atmospheric conditions, misalignment errors.	It includes errors due to displacement of level joints, errors due to friction.